

Currently Used Deprecated Functions Document

Revision

Version 5
2/25/26 8:36 PM

SME

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Abstract

This document describes the process of documenting in-use deprecated functions.

Group / Owner

Development / Software Developer

Motivation

This document is motivated by the need to track use of deprecated (unsafe) functions within safety-critical, cyber-physical systems.

Note: Within the context of this document, the terms ***security*** and ***cybersecurity*** are used interchangeably. It is presumed that the term ***security*** is being used in reference to ***cybersecurity*** and not ***physical security***.

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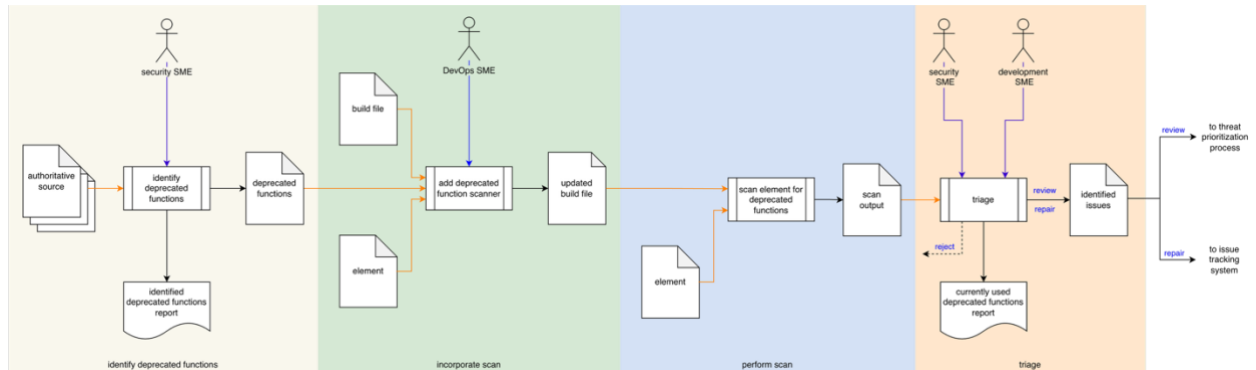
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Overview

Over time library functions may be determined to be insecure (unsafe). Where possible these should be replaced with secure alternatives. In order to assess the overall risk presented by those which cannot be replaced, they must be documented, and their continued use justified.

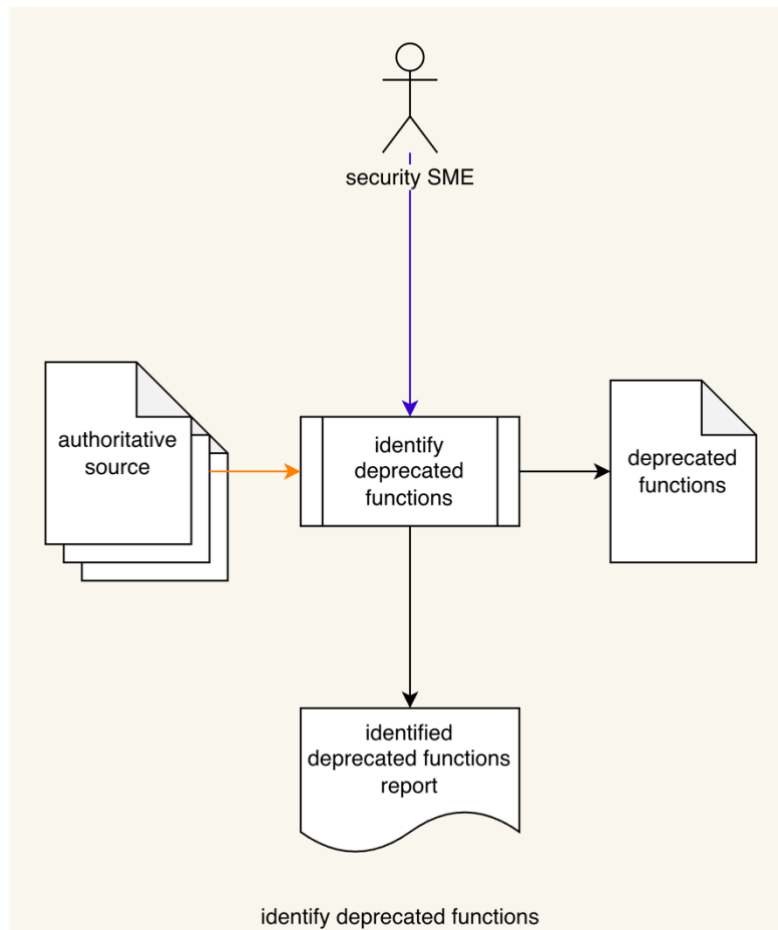
The following diagram illustrates the process to be used:



Process

Identify Deprecated Functions

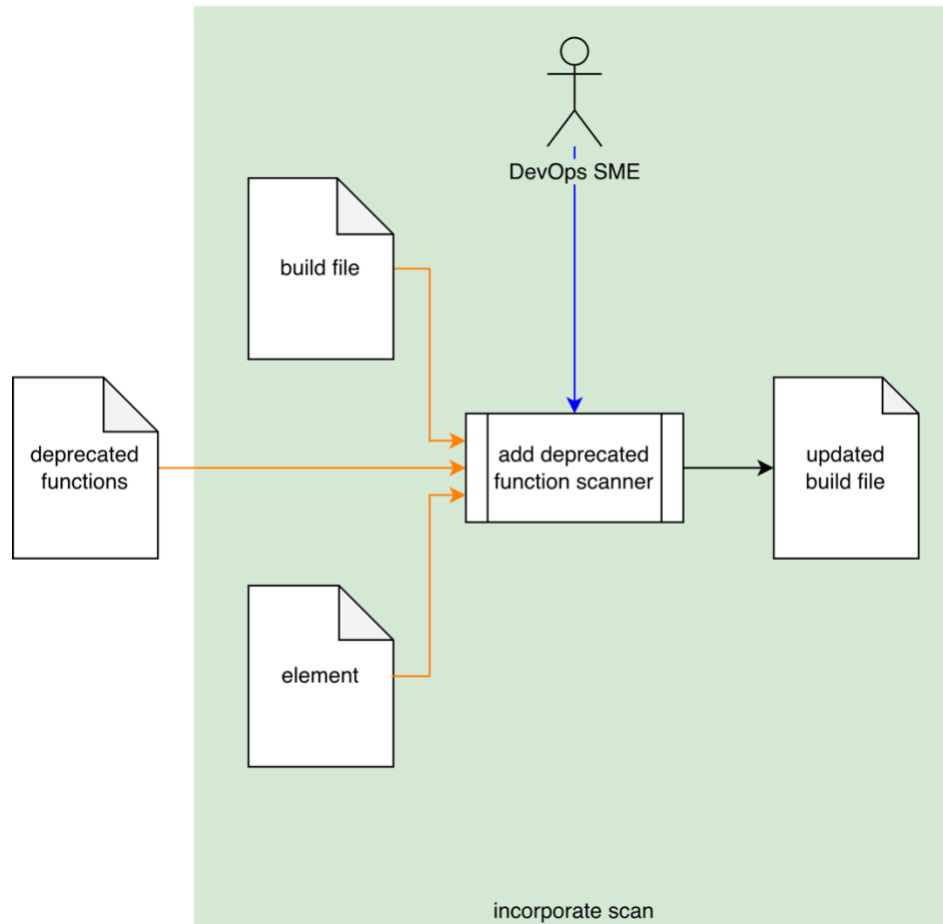
Inputs	Authoritative sources
Outputs	Deprecated functions Identified deprecated functions report
Participants	Security SME



The **Security SME** reviews the **Authoritative Sources** to identify functions which have been deprecated. These are entered into a **Deprecated Functions** list. An **Identified Deprecated Functions Report** is generated.

Incorporate Scan

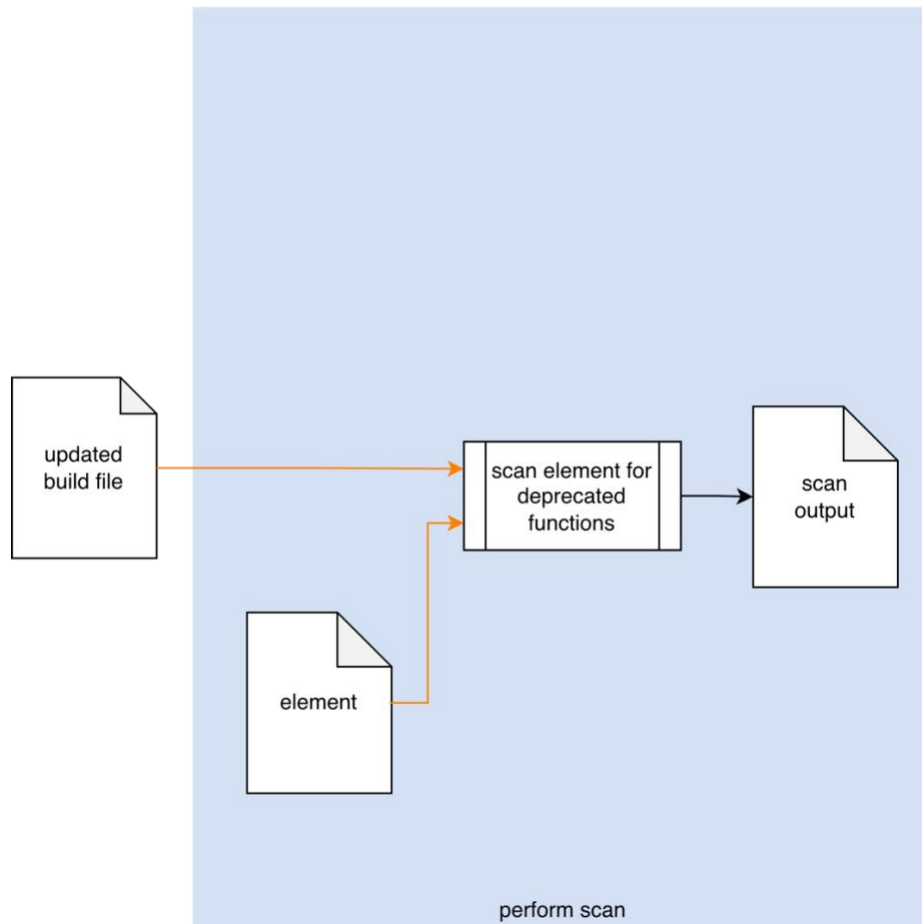
Inputs	Build file Element Deprecated functions
Outputs	Updated build file
Participants	DevOps SME



Using the **Deprecated Functions** list applicable to the **Element**, and **Build File**, the **DevOps SME** adds a function name scanner to the appropriate section of the **Build File**. An **Updated Build File** is created.

Perform Scan

Inputs	Updated build file
Outputs	Scan output
Participants	none



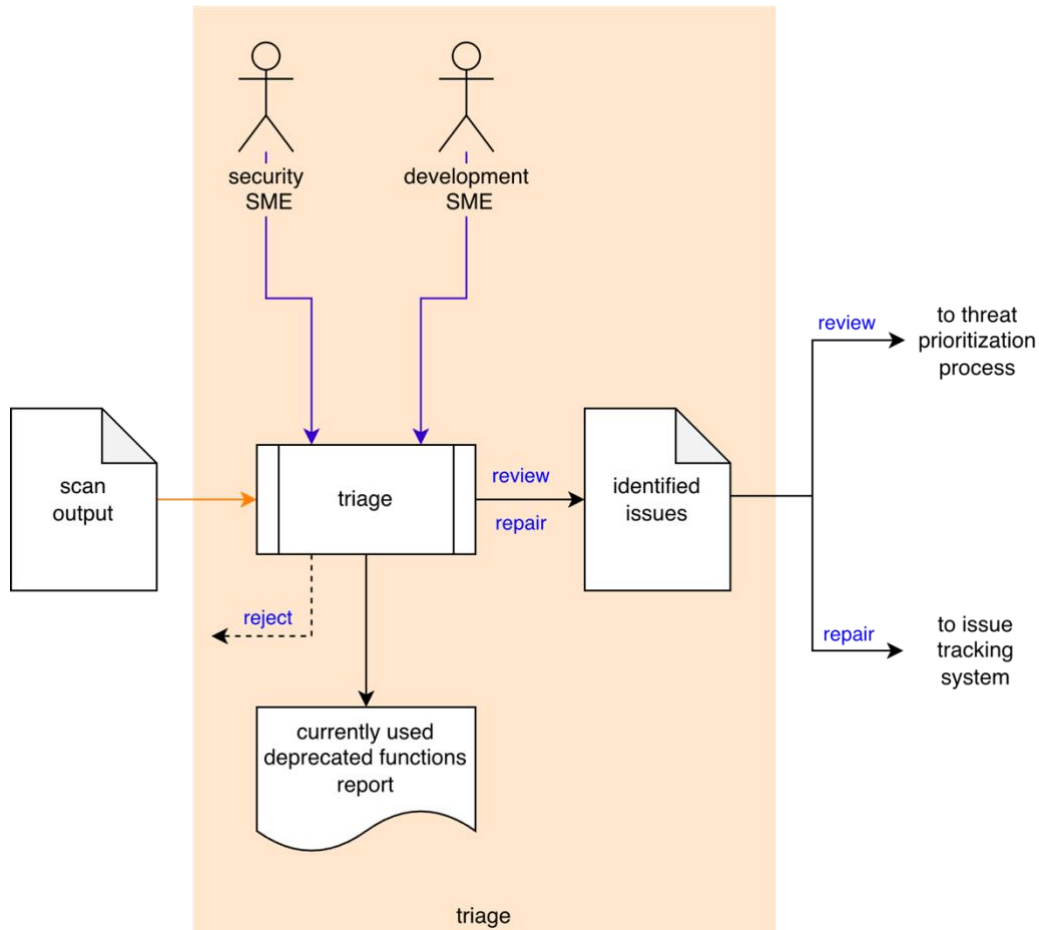
Using the **Updated Build File**, the **Element** is scanned for deprecated functions. A **Scan Output** list is generated.

Note: This activity may be performed manually.

Note: The recommended form of the **Scan Output** artifact is JSON-encoded Static Analysis Results Interchange Format (**SARIF**). This document assumes SARIF version 2.1.0 [\[2\]](#) or later.

Triage

Inputs	Scan output
Outputs	Identified issues Currently used deprecated functions report
Participants	Security SME Development SME



The **Security SME** and **Development SME** review the **Scan Output** to identify deprecated function usage needing further investigation. These **Identified Issues** are sent to the threat prioritization process [\[3\]](#). A **Currently Used Deprecated Functions Report** is be generated.

Note: Depending on the nature of an identified issue, it may be entered directly into the issue tracking system.

Note: The recommended form of the **Identified Issues** artifact is JSON-encoded **SARIF**.

Report

The **Currently Used Deprecated Functions Report** is recommended to be produced from the **Identified Issues** artifact and should detail the issues exposed by the static analysis.

The report contains one or more analysis runs. Each run includes:

- Description of the analyzed element
- Deprecated functions list
- Results of the analysis

The analyzed element description (one or more) provides information related to the element under consideration. Each includes:

- URI to analyzed element
- URI to repository the element came from

The deprecated functions (one or more) present. Each includes:

- ID (unique)
- Name
- Short description
- URI to function deprecation documentation
- Reference (optional) to associated taxonomy entry (Common Weakness Enumeration, ...)

The analysis results (one or more) describe the issues exposed by the analysis. Each includes:

- Human-readable description
- Location within the element of the issue
- Disposition of the issue

References

1. **AVCDL** (AVCDL primary document)
2. **Static Analysis Results Interchange Format (SARIF) Version 2.1.0**
<https://docs.oasis-open.org/sarif/sarif/v2.1.0/os/sarif-v2.1.0-os.pdf>
3. **Threat Prioritization Plan** (AVCDL secondary document)